Linear diff. eq. with constant coeff.

1. LHDE with CC.
2. Lm-HDE with CC.

1. LHDE with CC.

The characteristic eq. method

(1): $\boxed{x^{(m)} + a_1 \cdot x^{(m-1)} + \dots + a_{m-1} \cdot x' + a_m \cdot x = 0.}$, $m \in \mathbb{N}^*$, order of (1).

$a_1, \dots, a_m \in \mathbb{R}$ (the coeff. of (1)).

Step 1: Write the characteristic equation of (1):

$$\boxed{\ell(r) := r^m + a_1 \cdot r^{m-1} + a_2 \cdot r^{m-2} + \dots + a_{m-1} \cdot r^{m-(m-1)} + a_m = 0.} \quad (2)$$

(has degree m) and the same coeff. as (1).

Step 2: Find $r_1, \dots, r_m \in \mathbb{C}$ the m roots of (2).

↳ they can be repeated according to their multiplicities.

Step 3: We associate m functions, following the rules.

If $r_1 = \alpha \in \mathbb{R} \longmapsto e^{\alpha t}$
(simple)

If $r_{1,2} = \alpha \pm i \cdot \beta \longmapsto e^{\alpha t} \cdot \cos \beta t,\ e^{\alpha t} \cdot \sin \beta t.$
(simple)
$(\alpha, \beta \in \mathbb{R}, \beta > 0)$

If $r_1 = \alpha \in \mathbb{R}$ is a root of multiplicity $m \geq 2$:

$r_1 = \alpha \in \mathbb{R} \longmapsto e^{\alpha t}, t \cdot e^{\alpha t}, t^2 \cdot e^{\alpha t}, \dots, t^{m-1} \cdot e^{\alpha t}$. ⇒ m fct.

If $r_{1,2} = \alpha \pm i \cdot \beta$ is a root of multiplicity $m \geq 2$:
$(\alpha, \beta \in \mathbb{R}, \beta > 0)$ ⇒ 2m fct

$\longmapsto e^{\alpha t} \cdot \cos \beta t,\ e^{\alpha t} \cdot \sin \beta t,\ t \cdot e^{\alpha t} \cdot \cos \beta t, \dots, t^{m-1} \cdot e^{\alpha t} \cdot \sin \beta t$

In this way, we obtain m functions, let us denote them by $x_1, \dots, x_m$.

Step 4: Write the general solution of (1),

$$x = c_1 \cdot x_1 + \dots + c_m \cdot x_m, \quad c_1, \dots, c_m \in \mathbb{R}.$$

13.03.2026.

Lecture 3 – DS

Theorem 1: The m functions obtained at "Step 3" are solutions to the LHDE (1) and they are <u>linearly independent</u>.

<u>Notations:</u> $\mathcal{L}: C^m(\mathbb{R}) \to C(\mathbb{R})$.

$$\mathcal{L}(x) = x^{(m)} + a_1 \cdot x^{(m-1)} + \dots + a_m \cdot x.$$

$\Rightarrow$ (1) $\Leftrightarrow \mathcal{L}(x) = 0$.

$D: C^m(\mathbb{R}) \to C(\mathbb{R})$, $D(x) = x'$.

$\downarrow$

operator

$D \circ D = D^2$.

(compose)

$\underbrace{D \circ D \circ \dots \circ D}_{k \text{ times}} = D^k$.

$D^2(x) = D(D(x)) = D(x') = x''$.

$D^k(x) = x^{(k)}$.

$(a_1 \cdot D^k)(x) = a_1 \cdot x^{(k)}$

$a_1, \dots, a_m \in \mathbb{R}$.

$(D + a_m)(x) = x' + a_m \cdot x$

$$\begin{cases} r_1 + r_2 = r_2 + r_1 \\ r_1 \cdot r_2 = r_2 \cdot r_1 \end{cases}$$

$r_1, r_2 \in \mathbb{C}$.

$$(D - r_1) \circ (D - r_2) \overset{?}{=} D^2 - (r_1 + r_2) D + r_1 r_2 = (D - r_2)(D - r_1)$$

$$(D - r_1) \circ (D - r_2)(x) = (D - r_1)(x' - r_2 \cdot x) = D(x' - r_2 \cdot x) - r_1 \cdot (x' - r_2 \cdot x)$$

$$= x'' - r_2 \cdot x' - r_1 \cdot x' + r_1 \cdot r_2 \cdot x$$

$\downarrow$

"applied to a function x".

$\Rightarrow (D - r_1) \circ (D - r_2) = (D - r_1)(D - r_2)$.

<u>Lemma 1.</u> (The decomposition of $\mathcal{L}$).

Let $r_1, \dots, r_m$ the roots of the characteristic polynomial ℓ.

Then, $\ell(r) = (r - r_1) \cdot \ldots \cdot (r - r_m)$ and

$$\mathcal{L} = \ell(D) = (D - r_1) \cdot \ldots \cdot (D - r_m) \quad \text{(decomposition)}.$$

Proof.

$$l(r) = r^m + a_1 \cdot r^{m-1} + \dots + a_{m-1} \cdot r.$$

$$\mathcal{L}(x) = x^{(m)} + a_1 \cdot x^{(m-1)} + \dots + a_{m-1} \cdot x' + a_m \cdot x.$$

$$= D^{(m)}(x) + a_1 \cdot D^{(m-1)}(x) + \dots + a_{m-1} \cdot D(x) + a_m \cdot x.$$

$$= l(D)(x).$$

$$\Rightarrow \mathcal{L} = l(D).$$

Ex: Let $l(r) = (r-1)^2 \cdot (r+5)$

$$\mathcal{L} = (D-1)^2 \cdot (D+5)$$

$\mathcal{L}(x) = ?$ for $x \in \{e^{-t}, e^{-5t}, e^{t}, t \cdot e^{t}, t^2 \cdot e^{t}\}$.

- $\underline{x = e^{-t}}$: $(D+5)(e^{-t}) = (e^{-t})' + 5 \cdot e^{-t} = 4 \cdot e^{-t}$

$$(D-1)(e^{-t}) = (e^{-t})' - 1 \cdot e^{-t} = -2 \cdot e^{-t}$$

$$\mathcal{L}(e^{-t}) = (D-1)^2 \cdot (D+5)(e^{-t})$$
$$= (D-1)^2 \cdot (4 \cdot e^{-t})$$
$$= 4 \cdot (D-1)^2 (e^{-t})$$
$$= 4 (D-1)(-2 \cdot e^{-t})$$
$$= -8 (D-1)(e^{-t})$$
$$= -8 \cdot (-2 \cdot e^{-t})$$
$$= 16 \cdot e^{-t}$$

- $\underline{x = e^{-5t}}$: $(D+5)(e^{-5t}) = (e^{-5t})' + 5 \cdot e^{-5t} = 0$

$\mathcal{L}(e^{-5t}) = 0 \Rightarrow e^{-5t}$ is a solution of $\mathcal{L}(x) = 0$.

- $\underline{x = t \cdot e^{t}}$: $(D-1)(t \cdot e^{t}) = (t \cdot e^{t})' - t \cdot e^{t} = e^{t} + \cancel{t \cdot e^{t}} - \cancel{t \cdot e^{t}} = e^{t}$.

$$\mathcal{L}(e^{t}) = (D+5)(D-1)^2(e^{t}) = (D+5)(D-1)\underbrace{(D-1)(e^{t})}_{0} = 0$$

$\Rightarrow t \cdot e^{t}$ is a sol. of $\mathcal{L}(x) = 0$.

- $\underline{x = t^2 \cdot e^{t}}$: $(D-1)(t^2 \cdot e^{t}) = (t^2 \cdot e^{t})' - t^2 \cdot e^{t} =$

$$= 2t \cdot e^{t} + \cancel{t^2 \cdot e^{t}} - \cancel{t^2 \cdot e^{t}}$$
$$= 2t \cdot e^{t}.$$

$$\mathcal{L}(t^2 \cdot e^{t}) = (D-1)^2 (D+5)(t^2 \cdot e^{t}) =$$

$= (D+5)(D-1)(2t\cdot e^t)$

$= 2(D+5)(D-1)(t\cdot e^t)$

$= 2(D+5)(e^t)$

$= 2\left((e^t)' + 5\cdot e^t\right)$

$= 2\cdot 6\cdot e^t$

$= 12\cdot e^t.$

Lemma 2: Set $r_1 \in \mathbb{C}$. We have:

$$(D-r_1)^{k+1}\cdot(t^k\cdot e^{r_1\cdot t}) = 0,\ (\forall)\, k\in\mathbb{N}.$$

Proof: Induction.

$\underline{k=0}$: $(D-r_1)(e^{r_1\cdot t}) = 0$ true.

Assume for k: $(D-r_1)^{k+1}\cdot(t^k\cdot e^{r_1\cdot t}) = 0$.

Prove for $k+1$: $(D-r_1)^{k+2}\cdot(t^{k+1}\cdot e^{r_1\cdot t}) = 0$.

$(D-r_1)(t^{k+1}\cdot e^{r_1\cdot t}) = (t^{k+1}\cdot e^{r_1\cdot t})' - r_1\cdot t^{k+1}\cdot e^{r_1\cdot t}.$

$= (k+1)t^k\cdot e^{r_1\cdot t} + \cancel{r_1\cdot t^{k+1}\cdot e^{r_1\cdot t}} - \cancel{r_1\cdot t^{k+1}\cdot e^{r_1\cdot t}}$

$= (k+1)\cdot t^k\cdot e^{r_1\cdot t}$

$(D-r_1)^{k+2}\cdot(t^{k+1}\cdot e^{r_1\cdot t}) = (D-r_1)^{k+1}\cdot\left((k+1)\cdot t^k\cdot e^{r_1\cdot t}\right) \overset{\text{hyp.}}{=} 0$

Lemma 3:

a) If $\underline{r_1 = \alpha}$ is a root of multiplicity $\underline{m\geq 1}$ of ℓ, then $e^{\alpha t}, t\cdot e^{\alpha t}, \dots, t^{m-1}e^{\alpha t}$ are solutions to (1).

b) If $\underline{r_{1,2} = \alpha \pm i\cdot\beta}$ is a root of multiplicity $m\geq 1$ of ℓ, then $(\alpha,\beta\in\mathbb{R}, \beta>0)$

$e^{\alpha t}\cdot\cos\beta t,\ e^{\alpha t}\cdot\sin\beta t,\ t\cdot e^{\alpha t}\cos\beta t, \dots, t^{m-1}\cdot e^{\alpha t}\cdot\sin\beta t$ are sols. of (1).

Proof:

a) Let $r_1\in\mathbb{C}$. We want to prove that $\mathcal{L}(t^k\cdot e^{r_1\cdot t}) = 0$, $(\forall)\, k\in\{0,1,\dots,m-1\}$, provided that r_1 is a root of multiplicity $m\geq 1$ of ℓ.

$\ell(r) = \tilde{\ell}(r)(r-r_1)^m \longrightarrow \mathcal{L} = \tilde{\ell}(D)(D-r_1)^m$ $\underline{\underline{\text{Lemma 2}}}$

(decomposition)

$\Rightarrow k \in \{0,1,\dots,m-1\}$

$\Rightarrow k+1 \in \{1,2,\dots,m\}$

$\Rightarrow k+1 \leq m.$

$$\Rightarrow (D-r_1)^{m} \cdot (t^k \cdot e^{r_1 \cdot t}) = (D-r_1)^{\overbrace{m-(k+1)}^{\geq 0}} \cdot \underbrace{(D-r_1)^{k+1} \cdot (t^k \cdot e^{r_1 \cdot t})}_{0} = 0$$

$\Rightarrow L(t^k \cdot e^{r_1 \cdot t}) = 0.$

Take $r_1 = \alpha \in \mathbb{R}$. ...

b) Proof: $r_1 = \alpha + i \cdot \beta$.

$t^k \cdot e^{r_1 \cdot t}$, $k \in \{0,1,\dots,m-1\}$ are complex solutions.

? $\Rightarrow \mathrm{Re}(t^k \cdot e^{r_1 \cdot t})$ and $\mathrm{Im}(t^k \cdot e^{r_1 \cdot t})$ are real solutions.

$$t^k \cdot e^{r_1 \cdot t} = t^k \cdot e^{(\alpha + i\beta)t} = t^k \cdot e^{\alpha t} \cdot \cos\beta t + i \cdot t^k \cdot e^{\alpha t} \cdot \sin\beta t.$$

Prove lin. ind.

Lemma 4:

a) Let $r_1 = \alpha \in \mathbb{R}$ and $m \geq 1$.
Then $e^{\alpha t}, t \cdot e^{\alpha t}, \dots, t^{m-1} \cdot e^{\alpha t}$ are lin. ind.

b) Let $\alpha_1, \alpha_2, \dots, \alpha_m \in \mathbb{R}$.
$\alpha_i \neq \alpha_j$
Then $e^{\alpha_1 \cdot t}, \dots, e^{\alpha_m \cdot t}$ are lin. ind.

Proof

a) Let $c_1, \dots, c_m \in \mathbb{R}$ be s.t. $c_1 \cdot e^{\alpha t} + c_2 \cdot t \cdot e^{\alpha t} + \dots + c_m \cdot t^{m-1} \cdot e^{\alpha t} = 0$, $(\forall) t \in \mathbb{R}$.

$\Rightarrow \underbrace{e^{\alpha t}}_{>0}(c_1 + c_2 \cdot t + c_3 \cdot t^2 + \dots + c_m \cdot t^{m-1}) = 0$, $(\forall) t \in \mathbb{R}$. $| : e^{\alpha t}$

$\Rightarrow c_1 + c_2 \cdot t + c_3 \cdot t^2 + \dots + c_m \cdot t^{m-1} = 0$, $(\forall) t \in \mathbb{R}$

$\Rightarrow c_1 = c_2 = \dots = c_m = 0$, $(\forall) t \in \mathbb{R}$

b) Let $c_1, \dots, c_m \in \mathbb{R}$ s.t. $c_1 \cdot e^{\alpha_1 \cdot t} + c_2 \cdot e^{\alpha_2 \cdot t} + \dots + c_m \cdot e^{\alpha_m \cdot t} = 0 \,|'$, $(\forall) t \in \mathbb{R}$.

$\Rightarrow c_1 \cdot \alpha_1 \cdot e^{\alpha_1 \cdot t} + c_2 \cdot \alpha_2 \cdot e^{\alpha_2 \cdot t} + \dots + c_m \cdot \alpha_m \cdot e^{\alpha_m \cdot t} = 0 \,|'$

$\Rightarrow c_1 \cdot \alpha_1^2 \cdot e^{\alpha_1 \cdot t} + c_2 \cdot \alpha_2^2 \cdot e^{\alpha_2 \cdot t} + \dots + c_m \cdot \alpha_m^2 \cdot e^{\alpha_m \cdot t} = 0 \,|'$

...

$\Rightarrow c_1 \cdot \alpha_1^{m-1} \cdot e^{\alpha_1 \cdot t} + \dots + c_m \cdot \alpha_m^{m-1} \cdot e^{\alpha_m \cdot t} = 0$, $(\forall) t \in \mathbb{R}$.

$\underline{t=0}$.
$$\begin{cases} c_1 + \dots + c_m = 0 \\ c_1\cdot\alpha_1 + \dots + c_m\cdot\alpha_m = 0 \\ c_1\cdot\alpha_1^2 + \dots + c_m\cdot\alpha_m^2 = 0 \\ \dots \\ c_1\cdot\alpha_1^{m-1} + \dots + c_m\cdot\alpha_m^{m-1} = 0 \end{cases}$$

$$\Rightarrow W = \begin{vmatrix} 1 & \dots & 1 \\ \alpha_1 & \dots & \alpha_m \\ \alpha_1^2 & \dots & \alpha_m^2 \\ \dots & & \\ \alpha_1^{m-1} & \dots & \alpha_m^{m-1} \end{vmatrix}$$

$\underline{\text{Vandermonde}}$ $(\alpha_1-\alpha_2)(\alpha_1-\alpha_3)\cdot\ldots\cdot(\alpha_{m-1}-\alpha_m) \neq 0$

$\Rightarrow c_1 = c_2 = \ldots = c_m = 0$ (the only sol; unique).

Exercises:

1. Prove that $\{\sin t, \cos t, e^{2t}\}$ are lin.ind., using the $\underline{\text{definition}}$.
2. Prove that " — " — using Theorem 1.

$e^{2t} \longmapsto r_1 = 2$ is a root of the characteristic polynomial ℓ.

$\cos t, \sin t \longmapsto r = i$ is a root " — " $\underline{\quad}$.

$\ell(r) = (r-2)(r-i)(r+i)$.

$\Rightarrow \{\sin t, \cos t, e^{2t}\}$ are obtained using the char. eq. method for the char. polyn.

3. Consider $\mathcal{L}$ (general).

- Let x be a constant function. $\Big|$ $\Rightarrow \mathcal{L}(x)$ ct. function.
- Let $x = d\cdot e^{\alpha t}$. Then $\mathcal{L}(d\cdot e^{\alpha t}) = \underbrace{d\cdot\ell(\alpha)}_{ct}\cdot e^{\alpha t}$

$(D-r_1)(d\cdot e^{\alpha t}) = (d\cdot e^{\alpha t})' - r_1\cdot d\cdot e^{\alpha t} = d\cdot(\alpha - r_1)\cdot e^{\alpha t}$.

$1+i \Rightarrow \begin{matrix}\alpha = 1 \\ \beta = 1\end{matrix} \longmapsto e^t\cdot\cos t$.

$\alpha + i\cdot\beta \Rightarrow e^{\alpha t}\cdot\cos\beta t$

$e^{\alpha t}\cdot\sin\beta t$

$(\alpha = 0)$

2. Ln-HDE with CC.

Theorem 2. [The fundamental theorem].

The general solution of $\mathcal{L}(x) = f(t)$ is $\boxed{x = \underset{\text{gen.}}{x_h} + \underset{\text{part.}}{x_p}}$, where x_p is a particular solution of $\mathcal{L}(x) = f(t)$ and x_h is the general sol. of $\mathcal{L}(x) = 0$.

?

Proof:

x_p - particular sol. of $\mathcal{L}(x) = f(t) \Leftrightarrow \mathcal{L}(x_p) = f(t)$.

$\mathcal{L}(x) = f(t)$.

$\mathcal{L}(x - x_p) = \mathcal{L}(x) - \mathcal{L}(x_p) = f(t) - f(t) = 0.$

$\mathcal{L}$ linear!
(ind?)

$\Rightarrow x - x_p$ is a sol. of the LHDE associated.

Example: Find the general sol. of $x' - 2x = 5$. ← non-homog. part.

(this is a first-order linear, non-homog, diff. eq.) → Ln-HDE w CC.

$\Rightarrow \boxed{x = x_h + x_p.}$

particular sol. of $x' - 2x = \underline{5} \Rightarrow x_p = -\frac{5}{2}$.

general sol. of $x' - 2x = \underline{0} \Rightarrow x' - 2x = 0$

↓

$r - 2 = 0$

$r = 2 \mapsto e^{2t}$ (associate the function)

$x_h = c \cdot e^{2t}$ (add an arbitrary real ct.), $c \in \mathbb{R}$.

$\Rightarrow$ The general sol. of $x' - 2x = 5$ is $x = c \cdot e^{2t} - \frac{5}{2}$, $c \in \mathbb{R}$.

(x_h)

$\parallel$

$x' = 5 + 2x.$

$x_h = c \cdot e^{2t} - \frac{5}{2}$, $c \in \mathbb{R}$.